



UNIVERSITY OF SAN-JOSE RECOLETOS
Engineering Mathematical Society
A.Y. 2026 - 2027



Prepared by:
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“Trigonometric Functions”
Practice Problems

1. Find the Derivative of $f(x) = 3 \sin x + 4 \cos x$
2. Find $\frac{dy}{dx}$ if $y = 5 \tan x$
3. Differentiate: $y = 2 \sec x - 3 \csc x$
4. Find the Derivative of $f(x) = x^2 \sin x$
5. Find the Derivative of $f(x) = x^3 \sec x + 2 \sin x$
6. Find $\frac{dy}{dx}$ if $y = \sin(3x)$
7. Differentiate using the Quotient Rule: $y = \frac{\sin x}{x}$
8. Find the Derivative of $f(x) = (\sin x)^{21}$
9. Find the Derivative of $f(x) = (2 \cos x + 1)^5$ using the Chain Rule
10. Find the Derivative of $f(x) = x^3 \csc x + 2x^2 \cot x$



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“Trigonometric Functions”

Step-by-step Solution

1. Find the Derivative of $f(x) = 3 \sin x + 4 \cos x$

- **Rule:** Use the basic trigonometric derivative rules:

$$\triangleright \frac{d}{dx}(\sin x) = \cos x$$

$$\triangleright \frac{d}{dx}(\cos x) = -\sin x$$

- **Step 1:** Differentiate $3 \sin x$

$$\triangleright \frac{d}{dx}(3 \sin x) = 3 \cos x$$

- **Step 2:** Differentiate $4 \cos x$

$$\triangleright \frac{d}{dx}(4 \cos x) = -4 \sin x$$

- **Step 3:** Combine the derivatives:

$$\triangleright f'(x) = 3 \cos x - 4 \sin x$$

$$\text{Answer: } f'(x) = 3 \cos x - 4 \sin x$$

2. Find $\frac{dy}{dx}$ if $y = 5 \tan x$

- **Rule:** Use the derivative rule:

$$\triangleright \frac{d}{dx}(\tan x) = \sec^2 x$$

- **Step 1:** Keep the constant 5:

$$\triangleright \frac{dy}{dx} = 5 \frac{d}{dx}(\tan x)$$

- **Step 2:** Differentiate $\tan x$

$$\triangleright \frac{dy}{dx} = 5 \sec^2 x \quad ; \quad \text{Answer: } \frac{dy}{dx} = 5 \sec^2 x$$



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3. Differentiate: $y = 2 \sec x - 3 \csc x$

- **Rule:** Use the derivative rules:

$$\triangleright \frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\triangleright \frac{d}{dx}(\csc x) = -\csc x \cot x$$

- **Step 1:** Differentiate $2 \sec x$:

$$\triangleright \frac{d}{dx}(2 \sec x) = 2 \sec x \tan x$$

- **Step 2:** Differentiate $-3 \csc x$

$$\triangleright \frac{d}{dx}(-3 \csc x) = -3(-\csc x \cot x)$$

$$\triangleright \frac{d}{dx}(-3 \csc x) = 3 \csc x \cot x$$

- **Step 3:** Combine the derivatives:

$$\triangleright y' = 2 \sec x \tan x + 3 \csc x \cot x$$

$$\text{Answer: } y' = 2 \sec x \tan x + 3 \csc x \cot x$$

4. Find the Derivative of $f(x) = x^2 \sin x$

- **Rule:** Use the Product Rule:

$$\triangleright \frac{d}{dx}(uv) = u'v + uv'$$

- **Step 1:** Let:

$$\triangleright u = x^2$$

$$\triangleright v = \sin x$$

- **Step 2:** Find the derivatives:

$$\triangleright u' = 2x$$

$$\triangleright v' = \cos x$$



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- **Step 3:** Apply the Product Rule:

$$\triangleright f'(x) = (2x)(\sin x) + (x^2)(\cos x)$$

- **Step 4:** Simplify:

$$\triangleright f'(x) = 2x \sin x + x^2 \cos x$$

Answer: $f'(x) = 2x \sin x + x^2 \cos x$

5. Find the Derivative of $f(x) = x^3 \sec x + 2 \sin x$

- **Rule:** Use the Product Rule for $x^3 \sec x$

$$\triangleright \frac{d}{dx}(uv) = u'v + uv'$$

- **Step 2:** Remember its derivative rules

$$\triangleright \frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\triangleright \frac{d}{dx}(\sin x) = \cos x$$

- **Step 3:** Differentiate $x^3 \sec x$

$$\triangleright \frac{d}{dx}(x^3 \sec x) = (3x^2)(\sec x) + (x^3)(\sec x \tan x)$$

$$\triangleright \frac{d}{dx}(x^3 \sec x) = 3x^2 \sec x + x^3 \sec x \tan x$$

- **Step 4:** Differentiate $2 \sin x$

$$\triangleright \frac{d}{dx}(2 \sin x) = 2 \cos x$$

- **Step 5:** Combine the derivatives

$$\triangleright f'(x) = 3x^2 \sec x + x^3 \sec x \tan x + 2 \cos x$$

Answer: $f'(x) = 3x^2 \sec x + x^3 \sec x \tan x + 2 \cos x$



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6. Find $\frac{dy}{dx}$ if $y = \sin(3x)$

- **Rule:** Use the Chain Rule:

$$\triangleright \frac{d}{dx}(\sin u) = \cos u \frac{du}{dx}$$

- **Step 1:** Let:

$$\triangleright u = 3x$$

- **Step 2:** Differentiate the inner function

$$\triangleright \frac{du}{dx} = 3$$

- **Step 3:** Differentiate the outer function

$$\triangleright \frac{dy}{dx} = \cos(3x)(3)$$

- **Step 4:** Simplify:

$$\triangleright \frac{dy}{dx} = 3 \cos(3x)$$

Answer: $\frac{dy}{dx} = 3 \cos(3x)$

7. Differentiate using the Quotient Rule: $y = \frac{\sin x}{x}$

- **Rule:** Use the Quotient Rule:

$$\triangleright \frac{d}{dx}\left(\frac{u}{v}\right) = \frac{vu' - uv'}{v^2}$$

- **Step 1:** Let:

$$\triangleright u = \sin x$$

$$\triangleright v = x$$

- **Step 2:** Differentiate each:

$$\triangleright u' = \cos x$$

$$\triangleright v' = 1$$



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- **Step 3:** Substitute into the Quotient Rule:

$$\Rightarrow y' = \frac{(x)(\cos x) - (\sin x)(1)}{x^2}$$

- **Step 4:** Simplify

$$\Rightarrow y' = \frac{x \cos x - \sin x}{x^2}$$

Answer: $y' = \frac{x \cos x - \sin x}{x^2}$

8. Find the Derivative of $f(x) = (\sin x)^{21}$

- **Rule:** Use the Chain Rule:

$$\Rightarrow \frac{d}{dx}(u^n) = nu^{n-1} \frac{du}{dx}$$

- **Step 1:** Let:

$$\Rightarrow u = \sin x$$

- **Step 2:** Differentiate the outer function

$$\Rightarrow 21(\sin x)^{21-1} = 21(\sin x)^{20}$$

- **Step 3:** Differentiate the inner function

$$\Rightarrow \frac{d}{dx}(\sin x) = \cos x$$

- **Step 4:** Multiply:

$$\Rightarrow f'(x) = 21(\sin x)^{20} \cos x$$

Answer: $f'(x) = 21(\sin x)^{20} \cos x$

9. Find the Derivative of $f(x) = (2 \cos x + 1)^5$ using the Chain Rule

- **Rule:** Use the Chain Rule:

$$\Rightarrow \frac{d}{dx}(u^n) = nu^{n-1} \frac{du}{dx}$$



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- **Step 1:** Let:
 $\triangleright u = 2 \cos x + 1$
- **Step 2:** Differentiate the outer function
 $\triangleright 5 (2 \cos x + 1)^{5-1} = 5 (2 \cos x + 1)^4$
- **Step 3:** Differentiate the inner function
 $\triangleright \frac{d}{dx} (2 \cos x + 1) = -2 \sin x$
- **Step 4:** Multiply:
 $\triangleright f'(x) = 5 (2 \cos x + 1)^4 (-2 \sin x)$
- **Step 5:** Simplify:
 $\triangleright f'(x) = -10 \sin x (2 \cos x + 1)^4$

Answer: $f'(x) = -10 \sin x (2 \cos x + 1)^4$

10. Find the Derivative of $f(x) = x^3 \csc x + 2x^2 \cot x$

- **Rule:** Use the Product Rule:
 $\triangleright \frac{d}{dx} (uv) = u'v + uv'$
- **Step 1:** Remember its derivative function
 $\triangleright \frac{d}{dx} (\csc x) = -\csc x \cot x$
 $\triangleright \frac{d}{dx} (\cot x) = -\csc^2 x$
- **Step 2:** Differentiate $x^3 \csc x$:
 $\triangleright \frac{d}{dx} (x^3 \csc x) = (3x^2)(\csc x) + (x^3)(-\csc x \cot x)$
 $\triangleright \frac{d}{dx} (x^3 \csc x) = 3x^2 \csc x - x^3 \csc x \cot x$



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- **Step 3:** Differentiate $2x^2 \cot x$:

$$\rightarrow \frac{d}{dx}(2x^2 \cot x) = (4x)(\cot x) + (2x^2)(-\csc^2 x)$$

$$\rightarrow \frac{d}{dx}(2x^2 \cot x) = 4x \cot x - 2x^2 \csc^2 x$$

- **Step 4:** Combine the derivatives:

$$\rightarrow f'(x) = 3x^2 \csc x - x^3 \csc x \cot x + 4x \cot x - 2x^2 \csc^2 x$$

$$\text{Answer: } f'(x) = 3x^2 \csc x - x^3 \csc x \cot x + 4x \cot x - 2x^2 \csc^2 x$$